

Design Number: 3

Design Name: Designing PCB of Variable Gain Signal Conditioning Circuit

Station Number: 1

Submission Date: November 11, 2025

Objective

The objective of this design experiment was to familiarize ourselves with the process of printed circuit board (PCB) design and apply this to the variable gain signal conditioning circuit which we have had previous experience with.

Equipment

- (1) Computer with PSPICE Software

Procedure

Part 1: Circuit Simulation

1. Open ORCAD's PSPICE software by searching "Capture" in the Windows menu, and make a new file.
2. Construct the circuit of figure 1 below by dropping parts, connecting with wires, and assigning values. *Note: Import all libraries to gain access to the LM358 op-amp as well as importing the connector library which will be used later:

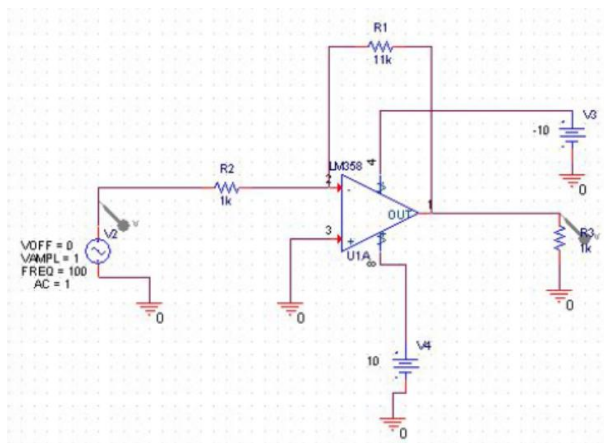


Figure 1: The following represents the signal conditioning circuit we will model our design procedure around.

3. Make a new simulation profile and apply differential voltage markers across the input and output.
4. Run the simulation. Observe and interpret the results. Record for later.

Part 2: PCB Design

1. Ensure the header library is imported as previously instructed.
2. Replace all voltage sources with jumpers from the header library.
3. Go to Edit >> Properties >> PCB Footprint. Add the footprint name: “Jumper2” for each header.
4. Go to Edit >> PCB >> Design Rule Check. Run the check, and click on “Run.”
5. Go to Tools >> Create Nestlist >> “Ok”
6. Go to PCB >> “New Layout” >> “OK”
7. From the “Design Parameters” set the x and y grid spacings to 25.00 each.
8. Set the “Shape” to “Rectangular”, and from “Board Geometry,” set W and H to 3000 and 2000 respectively.
9. From “Shape” >> “Z-copy” >> “Route Keepin” and “Package Keepin” >> “Offset”, set the offset to 100.00 and 250.00 respectively.
10. Drop all parts per “Manual Place” and connect per schematic.
11. Complete all checks and take a screenshot of the final layout.

Results

Part 1 Results

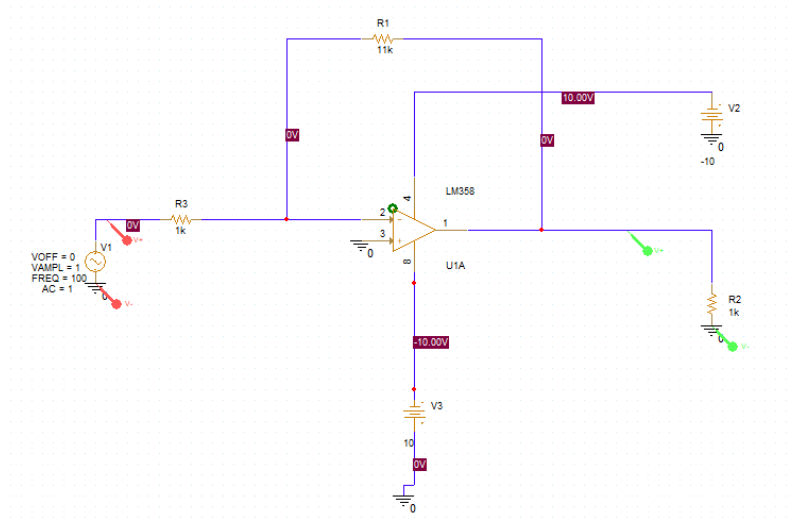


Figure 2: The following is the simulation of the voltage gain signal conditioning circuit of figure 1.

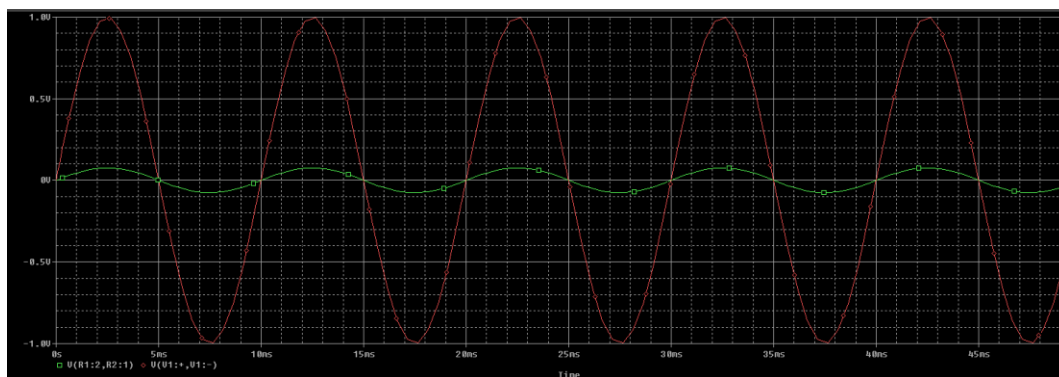


Figure 3: The following is the waveform of figure 2. The input signal is in green, and the output is in red.

Part 2 Results

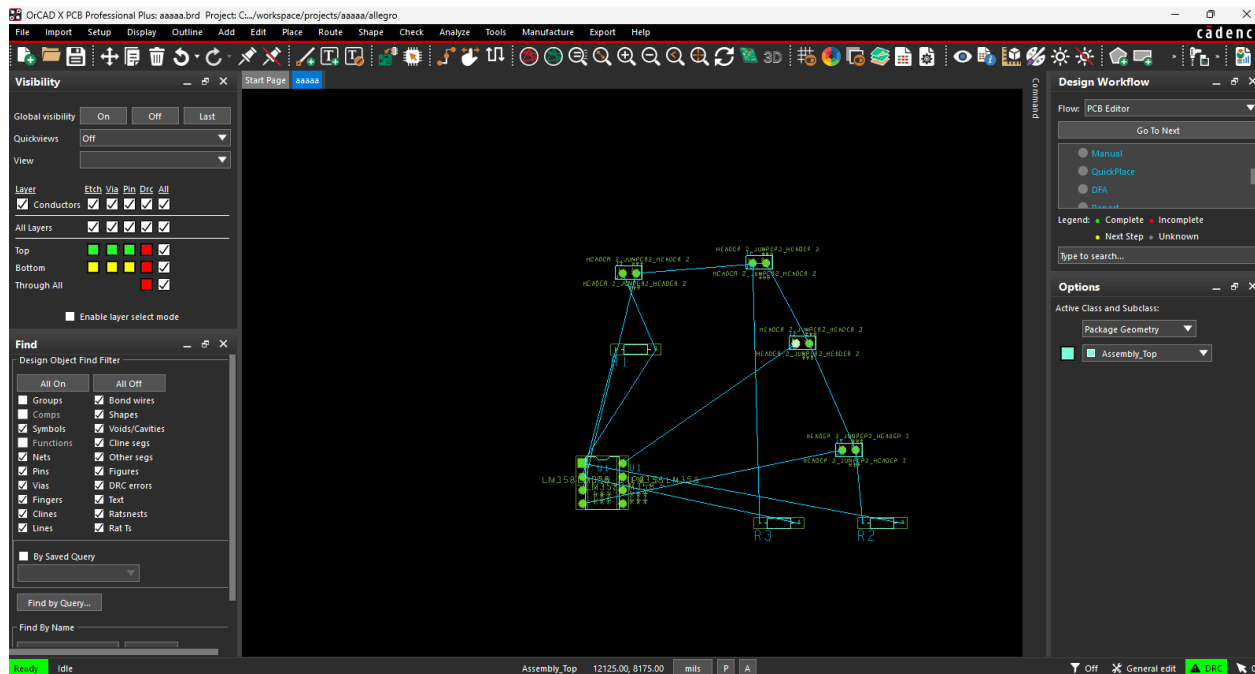


Figure 4: The following is the PCB Layout for part 2 of the procedure.

Report Questions

No report questions were provided for this design experiment.

Conclusion

In part 1 of this procedure, we confirmed our knowledge of the gain signaling current. We noticed that the output of the circuit was a scaled increase of the amplitude of our input signal while maintaining the other signal properties. The second part gave us our crude PCB design. It is worth noting that we were unable to achieve a properly “professional” looking PCB as this was our first attempt. However, we noticed that it maintained all the connections of the original simulated circuit. The main difference was the lack of voltage sources as these are replaced by power traces in common PCBs. In our case, we used headers as these would power the circuit external to the circuit board.

Preparation Problems

No preparation problems were provided for this design experiment.

Worksheet

Not applicable for this design experiment.